

Attachment 10

Airborne Gravity Gradiometry Technical Specifications

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1. General

- (a) This Schedule and its Attachments 1 to 9 form the Technical Specifications for airborne gravimetry and gravity gradiometry surveys.

2. Technical Glossary

- (a) The Technical Glossary is an addendum to, and forms a part of, the Contract Glossary.

Table 1 Technical Glossary

Acceptable Data	Data acquired that meet all specifications in this document.
Accuracy	As defined in ISO 5725-6:1994(en) and reproduced here in Attachment 3.
Aircraft	The aircraft used by the Supplier to conduct a Survey.
Aircraft Reference Point	The location in the aircraft that is used as the reference point for positional information. This will typically be the position of one of the major sensors.
Bias	As defined in ISO 5725-6:1994(en) and reproduced here in Attachment 3.
Commencement Report	A report documenting all pre-survey equipment calibrations and flight testing, in survey configuration, of the survey equipment.
Communications Plan	A standalone document or a portion of another document where the roles and responsibilities of the parties in terms of contacting stakeholders in the Survey Area and communications between the parties is described.
Coordinate Frame	The North, East, Down right-handed coordinate frame.
Daily Report	A report, provided daily, describing the operations and progress of acquisition.
Data	Any data acquired in the provision of the Services.
Data Filter Extension	an extension to a survey line beyond the survey boundary equal to the half-length of the longest low-pass filter used in processing the gravimetry point data. Data Filter Extensions contribute to the Chargeable Survey Distance.
Datum Alias	'GDA20'
Digital Terrain Model	The highest resolution model available of the elevation of the bare-earth relative to the Height Datum for comparison purposes with the Survey Elevation Model.
Field Data	Raw Data that have been manipulated or enhanced by processing to produce a dataset that is used during the acquisition phase of the survey by the Supplier and the Customer to perform QA/QC.
Final Data	Raw Data that have been manipulated or enhanced by processing to produce a final dataset.
Final Report	A detailed report on all aspects of acquisition and processing of the Raw Data acquired and processing for production of the final data.
Flight Path Plan	A digital record of the horizontal and vertical location of lines that are proposed to be flown as part of the survey. It includes flight lines, tie lines, Test Lines, and Repeat Lines. It includes the full extent of any line of data that will be delivered to the Customer — i.e., with filter data-padding extensions on either end of each line. It does not include lines flown as part of calibration activities.
Gravity Data	Data acquired by the airborne gravimeter or gravity gradiometer system used in the provision of the Services.
Gravity Datum	AAGD07 (Australian Absolute Gravity Datum 2007), Tracey et al, 2007.
Height Datum	Australian Height Datum (AHD)
Maximum Chargeable Survey Distance	The largest number of survey line-kilometres for which payment will be made under the Contract.
Precision	As defined in ISO 5725-6:1994(en) and reproduced here in Attachment 3.
Position Datum	Geocentric Datum of Australia 2020 (GDA2020)
Projected Coordinates	Map Grid of Australia (MGA)

Table 1 Technical Glossary

Projection Alias	'MGAXX' where XX is the zone number
Raw Data	Data as recorded digitally or in other form by the geophysical systems in the aircraft, by the geophysical systems on the ground, or by the Supplier's personnel.
Recording Resolution	As defined in ISO 5725-6:1994(en) and reproduced here in Attachment 3.
Repeat Line	A survey line that is flown a number of times by the Supplier so that the repeatability of the airborne gravity values can be determined. It may or may not coincide in horizontal location with ground gravity stations.
Re-flight Line Segment	a survey line or part thereof that is re-flown.
Risk Assessment	An airborne geophysical risk assessment based on the International Airborne Geophysics Safety Association (IAGSA) Safety Policy Manual (date of issue 01/12/2017) and Safety Management System (issue 01, 06/12/2018), These manuals can be accessed on https://www.iagsa.ca/members/publications
Run-in / Run-out	an extension to a survey line, beyond the Data Filter Extensions, required for instrument stabilisation. Run-in / Run-out extensions are not Chargeable.
Spatial Resolution	As defined in ISO 5725-6:1994(en) and reproduced here in Attachment 3.
Survey	The complete airborne geophysical project, encompassing all stages such as planning, acquisition, processing, and delivery.
Survey Elevation Model	Data recorded on the survey that are processed to represent the ground elevation relative to nominated vertical datum.
Trueness	As defined in ISO 5725-6:1994(en) and reproduced here in Attachment 3.
Weekly Report	A report, provided weekly, summarising the operations and progress of acquisition and the progress of data processing.

3. Coordinates and units

- (a) The Coordinate Frame must be used as the basis for the representation of all delivered vector or tensor components.
- (b) All geographic positions must be specified relative to the Position Datum; all projected coordinates to the Projected Coordinates and all heights to the Height Datum.
- (c) All measurements are to be reported in S.I units; in particular, the following units are to be used.

Table 2 Units

Gravimetry	μms^{-2} , $1 \mu\text{m/s}^2 = 10^{-6} \text{ ms}^{-2}$
Gravity Gradiometry	E: eotvos, $1 \text{ E} = 10^{-9} \text{ s}^{-2}$
Distance	m or km: metre or kilometre
Frequency	Hz: hertz, $1 \text{ Hz} = 1 \text{ s}^{-1}$ (unless radians per second are specified)

4. Field quality control and reporting

4.1. Quality control

- (a) It is the sole responsibility of the Supplier to demonstrate that the Data has been acquired in accordance with the Technical Specifications.
- (b) Re-flight Line Segments will be acquired at the expense of the Supplier if the Customer reasonably demonstrates that the Data do not meet the standards set out in the Technical Specifications.

- (c) Strict quality control of Data shall be maintained at all times with equipment calibration being checked at as required each day of flying.
- (d) The Supplier shall verify, on a daily basis, that the acquired data meets the Technical Specifications. It is expected that this will be either via an in-field data processing system set up at the Survey base, or the means to electronically send data to its processing office for verification. The Supplier will have a Data Verification Facility as specified in Attachment 7. The Data Verification Facility will be able to perform the following tasks at a minimum:
 - (i) identification and display of points where data is outside the Survey specifications;
 - (ii) display and verification of the flight path;
 - (iii) display and verification of the flying height;
 - (iv) calculation and reporting of statistics;
 - (v) identification and reporting of data that are not within the specifications;
 - (vi) checking for and editing of noise and spikes;
 - (vii) comprehensive display of data in profile form including repeat and calibration lines;
 - (viii) calculation and application of differential GNSS corrections;
 - (ix) calculation and application of parallax corrections;
 - (x) calculation and application of altimeter and ranging sensor corrections;
 - (xi) elevation data reductions including calculation of ground elevation;
 - (xii) gravimetric data processing including all corrections, filtering and levelling and tie-ing to the Gravity Datum;
 - (xiii) gravity gradiometric data processing including all corrections for aircraft dynamics and self-gradients, demodulation and filtering, transformation to all gravity gradient tensor components as well as gravity, terrain corrections and levelling;
 - (xiv) joint gravimetric and gravity gradiometric processing to produce optimal sets of gravity and gravity gradients based on the data from both sensor types;
 - (xv) gridding and on screen display of basic enhanced images of grids;
 - (xvi) exporting of data to the ASCII column format in accordance with Attachment 1 to enable transfer of digital quality control data to Customer's Representative;
 - (xvii) creation of data on portable hard drives or other suitable media to enable transfer of digital quality control data to Customer's Representative;
- (e) To verify that there are not any unexplainable spatial discontinuities in the dataset, the Supplier must produce and assess 'working grids' (grids of the data up to the stage of data acquisition). The working grids will include data selected by Customer's Representative.
- (f) Throughout acquisition, Customer's Representative shall be provided with a copy of the point located Field Data on regular intervals of 2 weeks. Requests for Field Data outside these intervals may also be made at any time by Customer.

4.2. Reporting

- (a) A Commencement Report must be supplied to the Customer prior to the commencement of the Survey and shall contain the information described in Section 11.3.
- (b) A Daily Report must be sent (typically by e-mail) to Customer's Representative within 24 hours of the reported days flying, or by the end of the following day if there was no flying, every day of the Survey from the start of acquisition until the end of acquisition and shall contain the information described in Section 11.4.
- (c) A Weekly Report must be supplied to the Customer weekly from the commencement of acquisition and until delivery of the Final Report and shall contain the information described in Section 11.5. A table of static data must be supplied with the Weekly Report as described in Attachment 1, Section 3.2.
- (d) A Final Report must be supplied to the Customer after the completion of all work on the Survey and shall contain the information described in Section 11.6.
- (e) In addition to the above requirements, 4.2(a) – (d), Customer's Representative must be notified:
 - (i) of any safety incidents; and

- (ii) before any equipment modifications or substitutions that may vary the quality of the Data; and
- (iii) if and when any Data are identified as being unusually noisy or not in accordance with the Technical Specifications.

5. Aircraft and equipment

5.1. Aircraft specifications

- (a) The Aircraft to be used is one of the aircraft specified in Attachment 5.
- (b) The Aircraft must conform to the following performance specifications at all times during the performance of the Services:
 - (i) The Aircraft must be approved by the Australian Civil Aviation and Safety Authority for the performance of geophysical survey operations.
 - (ii) The Aircraft performance must be sufficient to conduct the Survey safely over the Survey area at the nominal terrain clearance and at the nominal flying speed.
 - (iii) The Aircraft must have the necessary range and flight duration to efficiently fly the Survey.
 - (iv) The Aircraft speed in data acquisition configuration must be such that it can safely meet the minimum nominal spatial resolution specifications in Table A6.8 and/or Table A6.10.
 - (v) Climb rate performance must be sufficient to conduct the Survey over the Survey area at the nominal terrain clearance.
- (c) Customer approval must be obtained before an additional or replacement aircraft is introduced to the survey.
- (d) The Aircraft's nominal survey ground speed is specified in Attachment 5 (Table 5.3) of this Schedule. The mean speed over the entire survey will be close to this value as specified in Attachment 4 (Table A4.1), unless a larger variation is required for safety reasons. The nominal ground speed and the allowable variation must be agreed between the Customer and the Supplier before commencement of the Survey.
- (e) For any Survey to be conducted by the Supplier, the Customer may require the Supplier to undertake a formal safety audit of the aircraft operations prior to mobilisation of the survey aircraft to the survey area. The safety audit shall be performed by a licensed auditor at Customer expense. If the Supplier demonstrates that a safety audit has been conducted on the same aircraft within 12 months of the commencement of the survey, the Customer will check and accept the audit.

5.2. Equipment specifications

- (a) The Aircraft must be equipped with:
 - (i) a digital acquisition and recording system in accordance with Attachment 6;
 - (ii) a controlled fiducial numbering system to which all other instruments are to be accurately synchronised (i.e., for any instance all data channels recorded at that instance must be tagged with a unique fiducial for that line) in accordance with Attachment 6;
 - (iii) an integrated navigation system to measure aircraft flight path in three spatial dimensions and time in accordance with the requirements of Section 7 of this Schedule;
 - (iv) survey and base station GNSS equipment in accordance with Section 7 of this Schedule;
 - (v) aircraft to ground ranging equipment in accordance with Section 8 of this Schedule;
 - (vi) gravimetry equipment in accordance with Section 9 of this Schedule;
 - (vii) gravity gradiometry equipment in accordance with Section 10 of this Schedule;
 - (viii) other optional equipment (e.g., pressure gauge, thermometer, video flight path recorder) that may be specified in Attachment 6.

6. Survey Area

6.1. Survey area definition

- (a) The Request for Quotation provided by the Customer will contain locality maps outlining the area to be surveyed and tables of the specifications for the coverage of the Survey Area and coordinates of the boundary corners.
- (b) The Supplier will be responsible for the flight planning necessary to meet the coverage specifications of the Survey.
- (c) The proposed flight path and computed line lengths must be approved by the Customer before acquisition commences.
- (d) Material changes to the proposed flight path during the acquisition phase must be approved by the Customer before the changes are implemented.
- (e) The Supplier will ensure that all coordinate specifications will be referenced to the Position Datum. Coordinates may be specified in longitude and latitude (geodetic) coordinates, or in the relevant zone of the Map Grid of Australia (MGA) coordinates.
- (f) Geographic Information System (GIS) file(s) containing the start, intermediate waypoints (vertices where lines change direction), and end points of every line to be flown, will be attached to the Request for Quotation by the Supplier. The GIS file(s) will contain attributes for each survey line, including its number, and if applicable its sub-area name. The GIS files will be in either ESRI shape-files and/or MapInfo MID or MIF files.

6.2. Coverage specifications and tolerances

- (a) It is strongly emphasised by the Customer that the following items take precedence over any efforts to meet the flight specifications and tolerances:
 - (i) Pilot-determined safety considerations; and
 - (ii) The Australian Civil Aviation Safety Authority (CASA) rules governing low-level survey flying.
 - (iii) The recommendations from the Risk Assessment for the Survey that is submitted by the Supplier and accepted by the Customer.
- (b) Every effort within the bounds of safety considerations should be made to conduct the Survey at the planned altitude and on the planned survey lines.
- (c) Data acquired along lines that do not meet the specifications or tolerances will be Unacceptable Data.
- (d) The coverage specifications and tolerances are provided in Attachment 4.
- (e) Where two Aircraft are being utilised for data acquisition, each Aircraft must acquire data over separate contiguous blocks.
- (f) If contiguous flying is not possible, or is particularly inefficient, it should be reported to Customer's Representative prior to flying, and should be recorded in the Daily Report and Final Report.
- (g) When acquiring data the Supplier must make suitable run-in allowance in their flying pattern such that valid data are always acquired from the absolute start point to the absolute end point of each line. Thus, account must be made to ensure that the aircraft is on-track and flying straight and level (i.e., not in a turn) at the beginning to the end of the line. Account should also be made for any parallax shifts, filter taper lengths and stacking distances, that are to be applied during the data processing, which may mean processed data are truncated at the ends of lines.

6.3. Chargeable Survey Distance

- (a) The Maximum Chargeable Survey Distance for the Survey will be the total of the line lengths in the Customer-approved proposed Flight Path Plan. The lines in this flight plan include survey, tie, Test and Repeat lines. If gravimetry data is acquired, then each line has a Data Filter Extension on each ends, and these form part of the Flight Path Plan.
- (b) The Actual Chargeable Survey Distance for the Survey will be the aggregate length of the line segments from the Flight Path Plan along which Acceptable Data have been acquired and approved by the Customer.

- (c) Line segments that are re-flown to replace Unacceptable data will have extensions on either end of the portion of the line to be replaced. These are termed “overlaps”. The overlaps do not contribute to the Chargeable Survey Distance.
- (d) Data Filter Extensions are required for gravimetry data. These must be of sufficient length so that the Customer can re-process the line data with the same filters as used by the Supplier and obtain fully filtered data over the entire Survey Area. The Data Filter Extensions should be included in the planned lines in the Customer-approved proposed Flight Path Plan, and hence, will count towards Chargeable Survey Distance.
- (e) Payment under the contract will be made for the Actual Chargeable Survey Distance up to the limit of the Maximum Chargeable Survey Distance.

7. Navigation and flight path recording

- (a) All materials required for navigation are to be obtained by the Supplier.
- (b) The Navigation system specifications are to be as shown in Attachment 6.
- (c) The GNSS equipment (base station and rover) must be a dual-frequency 12 Channel GNSS receiver, or better.
- (d) Supplied positional Data are to be relative to the Position Datum. Any Data supplied in a Universal Transverse Mercator projection are to be in the Projected Coordinates.
- (e) Each flight line and each tie line is to have a unique number.
- (f) The line numbering system must be systematic and allow for easy identification and separation of flight lines, tie lines, repeat lines, and other calibration lines.
- (g) Navigation Data are to be accurately synchronised with the Data recorded from all other instruments using a fiducial marking system at a rate of at least 1 per second.
- (h) The following Data must be digitally recorded and supplied at the recording frequency specified in Table 3:
 - (i) longitude;
 - (ii) latitude;
 - (iii) GNSS height;
 - (iv) GNSS time;
 - (v) fiducial
- (i) The navigation and flight path recording system used must be capable of recording the aircraft position in flight to the specifications outlined in Table 3.

Table 3 Aircraft position in flight

Aircraft location recording frequency	≥ 1 Hz
Real-time horizontal position accuracy	≤ 3 m
Real-time vertical position accuracy	≤ 5 m

- (j) The number of satellites available to the GNSS system while the Aircraft is collecting data on line must be recorded and delivered. Data collected while the number of satellites is below the threshold in Table 4 are Unacceptable Data.
- (k) The PDOP, VDOP and HDOP values while the Aircraft is collecting data on line must be recorded and delivered. Data collected when any of these values is above the threshold in Table 4 below are Unacceptable Data.

Table 4 GNSS signal quality

Gravity survey mode	Minimum satellites	Maximum PDOP	Maximum VDOP & HDOP
gravimetry only	6	4	4
gradiometry only	5	4	4

Table 4 GNSS signal quality

joint gravimetry and gradiometry	6	4	4
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- (l) The GNSS Data must be post-processed using a GNSS base station surveyed in prior to commencement of the Survey.
- (m) The GNSS base station vertical and horizontal position must be accurate to better than 0.1 m.
- (n) The GNSS base station must be Surveyed in by either:
 - (i) using a known reference marker with accurate position and height, such as those that can be obtained from the National Mapping Division of Geoscience Australia, the details of which are to be provided in the Final Report. The reference marker to be used must have a horizontal and vertical accuracy of better than 0.1 metres. It should be noted that many trig marks and benchmarks do not have this accuracy. It is the responsibility of the Supplier to ensure that reference markers of appropriate accuracy are used; or
 - (ii) occupying the one location for the period of the Survey and averaging the height and positional data recorded by the GNSS.
- (o) The method used for surveying in the GNSS base station, the surveyed location and the estimated errors in the location must be described in the Final Report.
- (p) The final post-processing position accuracy must be better than 0.5 m RMS.
- (q) All GNSS Data will be post-processed by standard GNSS data post-processing software packages. The software packages used will be listed in the Final Report.
- (r) Real-time differential GNSS can be used in lieu of post-processing of GNSS data provided (i) a full constellation of satellites is visible and selective availability is not re-introduced by the developers of GNSS and (ii) the Supplier can demonstrate that the accuracy and precision of the calculated positions are as good as, or better than, than the accuracy and precision obtained by standard GNSS data post-processing methods.
- (s) At the beginning of each Survey flight, the Aircraft will be positioned at a suitable 'repeat-point' on the airport's apron.
 - (i) At the repeat-point data shall be recorded for sufficient time to allow recovery of a horizontal and vertical GNSS position. The GNSS data are then to be processed in the same way as for normal survey data.
 - (ii) The repeat-point horizontal location recovered from each flight must be within a 1 m radius of the repeat-point horizontal location recovered during the first flight from that airstrip.
 - (iii) The repeat-point GNSS height recovered from each flight must be within 2 m of the repeat-point GNSS height recovered during the first flight from that airstrip.
 - (iv) Repeat-point location results are to be plotted and tabulated to enable a comparison of processed positions accuracies for that day and previous days surveyed from that airstrip. The results shall be included in the Final Report.
- (t) The Supplier will take all reasonable steps to ensure the accuracy of final position recovery for all samples, and will fully disclose to the Customer the methods used and results of all quality control checks and calibration procedures carried out for this purpose.
- (u) It should be clear in the Final Report and in the metadata associated with the line data whether in-field or post-processed positional Data are used for any particular Data processing step, or whether the output of any Data processing step relates to in-field or post-processed positional Data.
- (v) The navigation system must enable lines to be flown on both geodetic and MGA coordinates. To clarify the distinction, this means that when, for example, flying an East-West line on geodetic coordinates a constant latitude would nominally be maintained, and when, for example, flying an East-West line on MGA coordinates a constant northing would nominally be maintained. In Attachment 1 of the Request for Quotation, Customer shall specify whether the survey lines are to be planned and flown according to geodetic or MGA coordinates.

8. Ranging sensors

8.1. General

- (a) A ranging sensor is required for the construction of the Survey Elevation Model used for terrain corrections of the gravimetric and / or gravity gradiometric data.
- (b) Ranging data may also be used to verify heights against the Height Datum and to check ground clearance.
- (c) Surveys incorporating gravity gradiometer measurements should include a laser scanner; otherwise a laser altimeter or radar altimeter may suffice. The choice of ground ranging equipment must be agreed by the Customer and specified in Attachment 6, Table A6.4.
- (d) The ranging sensor specifications must be sufficient to provide surface elevations to a trueness and precision such that the consequent terrain corrections can meet the requirements in Section 12.11.

8.2. Radar altimeter

- (a) The radar altimeter specified in Attachment 6, Table 6.4 will be used to measure the height of the Aircraft above ground level.
- (b) The radar altimeter will be calibrated according to the appropriate calibration procedure specified in Attachment 2.
- (c) The details of the calibration procedure and the consequent results will be described in the Final Report.

8.3. Laser altimeter

- (a) The laser altimeter(s) specified in Attachment 6, Table 6.4 will be used to measure the height of the Aircraft above ground level.
- (b) The laser altimeter will be calibrated according to the appropriate calibration procedure specified in Attachment 2.
- (c) The details of the calibration procedure and the consequent results will be described in the Final Report.
- (d) The laser altimeter will comply with Australian Government regulations with respect to the safe use of laser equipment.
- (e) The laser altimeter lens will be cleaned on a daily basis.

8.4. Laser scanner (LIDAR)

- (a) The laser scanner specified in Attachment 6.4 will be used to provide ranging data from the Aircraft over a swath of ground below the Aircraft.
- (b) The laser scanner will be calibrated according to the appropriate calibration procedure specified in Attachment 2.
- (c) The details of the calibration procedure and the consequent results will be described in the Final Report.
- (d) The laser scanner will comply with Australian Government regulations with respect to the safe use of laser equipment.
- (e) The laser scanner will be cleaned on a daily basis.

9. Gravimeter acquisition

- (a) If the Customer requires a gravimeter to be used for the Survey, then it will be the gravimeter specified in Attachment 6.
- (b) The gravimeter will be calibrated according to the appropriate calibration procedure specified in Attachment 2.
- (c) The details of the calibration procedure(s) and the consequent results will be described in the Final Report.
- (d) Any gravimeter data that does not meet the specifications in Section 6 of Attachment 6 is Unacceptable Data.

10. Gravity gradiometer acquisition

- (a) If the Customer requires a gravity gradiometer to be used for the Survey, then it will be the gravity gradiometer specified in Attachment 6.
- (b) The gravity gradiometer will be calibrated according to the appropriate calibration procedure specified in Attachment 2.
- (c) The details of the calibration procedure(s) and the consequent results will be described in the Final Report.
- (d) Any gravity gradiometer data that does not meet the specifications in Section 5 of Attachment 6 is Unacceptable Data.

11. Deliverables

11.1. Field Data

- (a) Field Data should be delivered to the Customer's Representative for QA / QC typically weekly throughout acquisition.
- (b) Field Data must at least be checked and edited in accordance with Section 12.3 of this Schedule before being delivered.
- (c) Field Data must be parallax corrected in accordance with Section 12.4 of this Schedule before being delivered.
- (d) The Field Data that must be supplied and the format in which they are to be supplied is set out in Attachment 1.
- (e) At the completion of acquisition, a complete set of Field Data for the entire survey shall be delivered to the Customer's Representative.

11.2. Final Data

- (a) Final Data must be fully processed according to all the requirements of Section 12 of this Schedule before being delivered.
- (b) Final Data must be supplied as specified in Attachment 1.

11.3. Commencement Report

- (a) The Commencement Report shall be provided by the Supplier to the Customer prior to commencement of survey acquisition.
- (b) The Commencement Report will describe in detail all required tests and calibrations performed in preparation for the Survey. If a required test or calibration was performed but details cannot be provided then reasons why details cannot be provided must be stated, the date of performance must be given, and whether or not the outcome was successful must be stated. The following tests and calibrations are required:
 - (i) instrumentation lag measurements for all sensors / instruments
 - (ii) radar altimeter calibration
 - (iii) laser altimeter calibration
 - (iv) laser scanner calibration
 - (v) gravity gradiometer self-gradient calibration
 - (vi) all gravity gradiometer calibrations
 - (vii) gravimeter calibrations
 - (viii) verification of navigation system to test the accuracy of the GNSS systems.
- (c) The Commencement Report will describe in detail the acquisition over a gravity test range and will report on the results.

11.4. Daily Reports

- (a) The Daily Report shall contain at least the following information:

- (i) The date reported on, the Customer Project identifier, the Supplier project identifier, the Survey name, the block name if there is more than one survey block, the base of operations, aircraft registration(s);
 - (ii) A list of personnel on site and their roles;
 - (iii) A summary of HSE activities and events, including any safety incidents;
 - (iv) A table of all lines flown and relevant statistics, including line length, mean speed, mean vertical turbulence, and mean error or noise in each gravity and gradiometer sensor
 - (v) the number of line kilometres flown today and accepted;
 - (vi) the number of line kilometres flown today and not accepted;
 - (vii) the total number of line kilometres flown and accepted to date;
 - (viii) the percentage of each block completed;
 - (ix) the total number of line kilometres remaining;
 - (x) weather conditions;
 - (xi) flight numbers, times and duration;
 - (xii) standby claims and reason;
 - (xiii) A flight plan map showing all lines and clearly indicating which lines are planned, which have been flown today, which have been flown previously, which have been approved by QC, and which require re-flight.
 - (xiv) results of pre- and post-flight gravimeter static measurements and pre- and post-gravity gradiometer Quiescent Noise measurements;
 - (xv) whether or not any maintenance was done on the aircraft;
 - (xvi) whether or not there were any problems with, or alterations to, any equipment;
 - (xvii) for each survey line and each static or quiescent test, a note to show if any data are out of specification;
 - (xviii) for any gravimetry repeat line flown, a note to show if data are out of specification;
 - (xix) the coordinates of the location of areas flown, on the day reported, for equipment calibration and compensation.
- (b) Plots must be in a suitable form, at appropriate scales and suitably annotated to allow adequate examination of the data for quality control purposes.

11.5. Weekly Reports

- (a) During acquisition, a weekly report summarising progress of the Data acquisition must be sent to Customer weekly. The report can be via email, and must reach Customer's Representative within 24 hours of the reported weeks flying.
- (b) Once acquisition is complete, weekly reports regarding the status of data processing are required. These can report on the status of the data processing only.

11.6. Final Report

- (a) The Supplier is required to provide a comprehensive and accurate Final Report on the survey operations and data processing.
- (b) The Final Report must be delivered in digital format as either a Microsoft Word document or a Portable Document Format (PDF) file.
- (c) The Final Report should be formatted for A4 (29.7 x 21 cm) paper size. The text should be neatly presented in a fully justified 16 cm wide single column and all accompanying maps, figures, etc. should be neatly presented and appropriately titled/labelled.
- (d) Digital draft copies of the Final Report will be supplied to Customer's Representative for comment and approval prior to the final delivery.
- (e) The report will include but not be restricted to details of:
 - (i) Survey location (schematic flight path locality map), coverage specifications, dates, and base(s) of operations;

- (ii) personnel;
- (iii) line kilometres flown;
- (iv) flight logs, standby claimed and reasons;
- (v) aircraft and equipment used;
- (vi) summary tables and graphs of calibration flights and other test results, including altimeter linearity tests, parallax tests, gravimeter and gravity gradiometer static measurements, repeat line comparisons, and laser altimeter and scanner pointing offsets and other constants;
- (vii) navigation report giving methods and operational details;
- (viii) equipment and systems used in Data processing;
- (ix) comprehensive processing and presentation procedures and results of Data corrections;
- (x) summary of the supplied Data and description of the formats;
- (xi) the coordinates of the location of areas flown for equipment calibration and compensation during the acquisition phase.

11.7. Delivery

- (a) All Data must be accurately and definitively annotated before being delivered, and must be accompanied by a plain text README file describing the contents.
- (b) All Data and deliverables are to be delivered to Customer's Representative by a secure digital method agreed between the Supplier and the Customer.

12. Data processing

12.1. Quality control of processed Data

- (a) It is the sole responsibility of the Supplier to ensure that the Data are processed in accordance with the Technical Specifications.
- (b) If the Supplier is not carrying out the work to the standard documented in this Deed, Customer's Representative may inform the Supplier that the Data are unacceptable and may require the Data to be re-flown or reprocessed at the Supplier's expense.
- (c) Regular Weekly Reports must be submitted during the processing phase to document progress towards delivery of the final products from the Survey.
- (d) Micro-levelling, if necessary, is to be done such that the micro-levelling is reflected in the located data and not just the grids. Where micro-levelling is undertaken, the located data will contain unlevelled, levelled and micro-levelled data fields.

12.2. Reporting of processing

- (a) The full details of the processing performed on all data, including any corrections, terrain corrections, levelling, micro-levelling, and all the details of each filter applied to the data, must be provided in the Final Report.

12.3. Checking and editing

- (a) Checking and, where necessary, editing of digital data is to be carried out as follows:
 - (i) removal of spikes;
 - (ii) removal of base level shifts; and
 - (iii) flagging of Data gaps as specified in Attachment 4.

12.4. Parallax corrections

- (a) Parallax and lag correction must be applied to account for any timing lags or cable length parallaxes between the different geophysical and GNSS instruments.
- (b) Should it be necessary to apply a parallax correction to the geophysical Data, the navigation positions must be interpolated rather than interpolating the geophysical Data to coincide with the navigation.

12.5. Navigation and flight path processing

- (a) The Supplier will take all reasonable steps to ensure the accuracy of final position recovery for all samples, and will fully disclose to the Customer the methods used and results of all quality control checks and calibration procedures carried out for this purpose.
- (b) The position information must relate to the Aircraft Reference Point.
- (c) The choice of Aircraft Reference Point must be approved by the Customer.
- (d) Measurements of horizontal location, elevation, and terrain clearance must be corrected for offsets between the relevant sensors and the Aircraft Reference Point.
- (e) Post-processed navigation Data which are linked to the geophysical Data from the other systems must be adjusted for any timing lags.
- (f) Final positional Data are to be supplied using the Position Datum and Projected Coordinates.
- (g) It should be clear in the Final Report and in the metadata associated with the line data whether in-field or post-processed positional Data are used for any particular Data processing step, or whether the output of any Data processing step relates to in-field or post-processed positional Data.

12.6. Laser altimeter data processing

- (a) The following corrections and compensations are to be applied to the raw laser range Data;
 - (i) removal of spurious returns or dropouts,
 - (ii) application of local maximum filters to recover “bare earth” solutions (i.e. to remove effects of noise spikes and above ground features such as vegetation and buildings), and
 - (iii) infill the gaps created in steps (1) and (2) above with spline interpolation.
- (b) The laser altimeter calibration constants derived through the calibration procedure specified in Attachment 2 (Section 2.2) will be used to correct the laser altimeter data for the effect of the laser altimeter’s pointing direction. The following equation will be used, where, l_c and l_m are the corrected and raw measured altimeter values respectively, p_m and r_m are

$$l_c = l_m \cos(p_m + p_o) \cos(r_m + r_o) + d_o \sin(p_m + p_o)$$

the measured pitch and roll of the laser altimeter’s mounting respectively, p_o and r_o are the pitch and roll pointing offset calibration constants respectively, and d_o is the horizontal offset between the laser altimeter and the centre of rotation of its’ mounting.

The Supplier may use alternative procedures to the above if agreed by the customer. The Supplier should then provide details of the procedures in the commencement report.

12.7. Laser scanner data processing

- (a) The laser scanner data will be calibrated according to the procedure specified in Attachment 2 (Section 2.3).
- (b) The laser scanner data will undergo the following processing steps:
 - (i) removal of spurious returns or drop-outs;
 - (ii) application of local maximum filters to recover “bare earth” solutions (i.e. to remove effects of noise spikes and above ground features such as vegetation and buildings), and
 - (iii) conversion from range data to ground elevations using the calibration data and aircraft attitude information from the gravity gradiometer or other suitable angular sensor(s);
 - (iv) gridding the ground elevations to produce a Survey Elevation Model;
 - (v) merging the Survey Elevation Model with other digital elevation gridded data to produce a final Digital Terrain Model;
 - (vi) the final Digital Terrain Model must have a grid cell size no greater than one-third of the ground clearance of the Aircraft, and a vertical accuracy of 0.5 m or better.

12.8. Gravimetric Data processing

- (a) The raw gravity data will be processed as specified in Attachment 8 or as otherwise agreed by the Customer.

- (b) The terrain model used for gravity or gravity gradient terrain corrections should be the best available Digital Terrain Model. The Survey Elevation Model should be used within the Survey Area if it is better than the pre-survey Digital Terrain Model in which case, the Digital Terrain Model outside the Survey Area should be calibrated with the Survey Elevation Model.
- (c) For each gravity-related field/column in the line data files, it must be clear whether these data relate in position to observations at their measured locations or whether they relate in position to observations continued onto a smooth drape surface.
- (d) If any gravity-related fields/columns in the line data files relate in positions to observations on a smooth drape surface, the smooth drape surface must be supplied as a grid and as a column in the line data files.
- (e) Final data should be provided in the Gravity Datum.

12.9. Gravity Gradiometric Data processing

- (a) The raw gravity gradient data will be processed as specified in Attachment 8 or as otherwise agreed by the Customer.
- (b) The processing is expected to include the following steps:
 - (i) removal of residual signal due to aircraft dynamic effects;
 - (ii) demodulation and filtering by a Demodulation Filter with characteristics specified in Table A6.8;
 - (iii) correction for Aircraft and platform self-gradients of gravity;
 - (iv) correction for any drift and base-level offsets;
 - (v) correction for terrain gravity gradients using a terrain density selected by analysis of the data or by a value supplied by the Customer;
 - (vi) tie-line levelling;
 - (vii) transformation to vertical gravity and to all components of the gravity gradient tensor in their Coordinate Frame representations.
- (c) The terrain model used for gravity or gravity gradient terrain corrections should be the best available Digital Terrain Model. The Survey Elevation Model should be used within the Survey Area if it is better than the pre-survey Digital Terrain Model in which case, the Digital Terrain Model outside the Survey Area should be calibrated with the Survey Elevation Model.
- (d) For each gravity-related field/column in the line data files, it must be clear whether these data relate in position to observations at their measured locations or whether they relate in position to observations continued onto a smooth drape surface.
- (e) If any gravity-related fields/columns in the line data files relate in positions to observations on a smooth drape surface, the smooth drape surface must be supplied as a grid and as a column in the line data files.

12.10. Joint Gravity Data processing

- (a) Data consisting of vertical gravity and gravity gradient tensor components all generated by the joint processing of gravimetric and gravity gradiometric data are Joint Gravity Data.
- (b) Where specified by the Customer, the Supplier is required to also deliver Joint Gravity Data.
 - (i) The Supplier may use externally available gravimetric and / or gravity gradiometric data in their joint processing.
- (c) Where the Survey includes acquisition of only gravity gradiometry, the Customer may require the Supplier to also deliver Joint Gravity Data using externally supplied gravimetry.
- (d) The Supplier must specify in Attachment 8, Section 6 how the gravimeter and gravity gradiometer data are to be merged including:
 - (i) full details of any filtering and / or gridding;
 - (ii) full details of how the merged data are tied to a gravity datum / network;
 - (iii) full details of any externally available gravimetric or gravity gradiometric data used in the processing.
- (e) The merged data ought to be delivered as a separate database and separate set of grids.

12.11. DTM data processing

- (a) A DTM is required for terrain corrections of gravity and gravity gradiometry data. Terrain corrections based on the DTM are required to be more accurate than the gravity and gravity gradiometer data. In particular:
 - (i) the DTM will be a digital file of bare–earth elevations over a regular grid of x and y positions;
 - (ii) the x and y positions will be in the Projected Coordinates and the elevations will be relative to the Height Datum;
 - (iii) the grid cell size will be less than one–third of the nominal ground clearance and sufficiently small as to introduce no visible artefacts in a grid of any terrain correction channel;
 - (iv) the elevations below the flown survey lines will be accurate to better than 0.5 m for gravity gradiometric terrain corrections and better than 5 m for gravimetric terrain corrections unless agreed otherwise by the Customer;
 - (v) the DTM will fully cover an area that extends beyond the Survey Boundary for at least 60 km for gravity gradiometric terrain corrections and 166.7 km for gravimetric terrain corrections unless agreed otherwise by the Customer.
- (b) The DTM may be generated by combining bare–earth elevations calculated from ranging sensor(s) during the survey (the Survey Elevation Model) and other pre–existing elevation data. If a Survey Elevation Model is used:
 - (i) it must be generated according to the procedure described in Attachment 8 Section 3.
- (c) The DTM may be micro-levelled where appropriate.
- (d) The final DTM must be checked against existing topographic maps and spot heights, to ensure consistency of the datasets within GNSS and ranging accuracy.
- (e) The DTM should be of such a quality that no streaking is evident on enhanced images of the Data with artificial illumination applied in a direction perpendicular to the flight line direction.

12.12. Gridding

- (a) Located Data are to be gridded to a grid cell size of one fifth of the minimum line spacing or as otherwise agreed by the Customer. If a Survey has variable line spacings, this will require the use of a variable density gridding technique (Murray, 1998).

13. Continual assessment

- (a) Throughout the term of this Deed, the Customer shall continue to assess the Supplier's airborne gravity system. This will allow the Supplier the opportunity to supply to the Customer the most up-to-date system specifications and sample data to demonstrate its system's capabilities and noise characteristics. This information will be used by the Customer when it assesses the technical suitability of each system on the Panel to carry out a specific survey.
- (b) The Supplier may supply up-to-date system specifications including:
 - (i) new or improved system hardware;
 - (ii) new or improved processing software or procedures;
 - (iii) sample raw and processed survey data; and
 - (iv) sample raw and processed repeat line data.
- (c) System noise will be assessed by the Customer on the basis of sample survey data collected over an airborne gravity test range such as the Kauring Gravity Test Range at Kauring, Western Australia (<http://www.ga.gov.au/about/projects/resources/kauring>);
- (d) The Customer will use the noise characteristics (Table A6.8 for gravity gradiometry and Table A6.10 for gravimetry) in its assessment of the Supplier's system's technical suitability for a specific survey.
- (e) Until such time as updated sample data are provided by the Supplier, and then new noise characteristics are agreed upon by the Supplier and the Customer, the allowable noise levels are those specified in the Noise Characteristics tables (Table A6.8 in Attachment 6.5.2 for gravity gradiometry and Table A6.10 in Attachment 6.6.2 for gravimetry).

14. References

- (a) Australian Society of Exploration Geophysicists Technical Standards Committee: ASEG-GDF2 data transfer standard, <https://www.aseg.org.au/technical/aseg-technical-standards>.
- (b) Brewster, J. 2016, Comparison of gravity gradiometer designs using the 3D sensitivity function, In SEG Technical Program Expanded Abstracts, Society of Exploration Geophysicists.
- (c) Difrancesco, D., 2001, Advances In Gravity Gradiometer Instrumentation At Lockheed Martin, presented at Gravity Gradiometer Workshop, SEG Conference, San Antonio, September 2001.
- (d) Hinze, W. J., et al., 2005. New standards for reducing gravity data: The North American gravity database. *Geophysics*, vol. 70, no. 4 (July-August 2005); p. j25–j32.
- (e) Howard, D., M. Gujic, and R. Lane, 2010, The Karing airborne gravity and airborne gravity gradiometer test site, Western Australia: in R.J.L. Lane (editor), *Airborne Gravity 2010 Abstracts from the ASEG-PESA Airborne Gravity 2010 Workshop*: Published jointly by Geoscience Australia and the Geological Survey of New South Wales, Geoscience Australia Record 2010/23 and GSNSW File GS2010/0457, pp. 107-114.
- (f) ISO, 1994, Standard ISO 5725-1:1994(en), Accuracy (trueness and precision) of measurement methods and results — Part 1: General principles and definitions. International Organisation for Standardisation, Geneva. Retrieved from: <https://www.iso.org/obp/ui/#iso:std:iso:5725:-1:ed-1:v1:en>.
- (g) Murray, A., 1998. High precision gridding of gravity data. 13th ASEG Conference, Extended Abstracts.
- (h) International Airborne Geophysics Safety Association (IAGSA) Safety Policy Manual. Available from IAGSA (<https://www.iagsa.ca>).
- (i) Tracey, R., M. Bacchin, and P. Wynne, 2007, AAGD07: A new absolute gravity datum for Australian gravity and new standards for the Australian National Gravity Database: 2007 ASEG-PESA 19th International Geophysical Conference and Exhibition, Extended Abstracts: <http://library.seg.org/doi/abs/10.1071/ASEG2007ab149>

Attachments to the Technical Specifications Schedule

Attachment 1 - Processed Digital Data

A1.1 - Point located data format

- (a) Point located data are to be fully compliant with the ASEG-GDF2 data standard, and shall be accompanied by the ASEG-GDF2 appropriate definition header (.dfn), description (.des) and projection (.prj) files.
- (b) Point located data are to be delivered in ASCII column format. The format must conform to the following criteria;
 - (i) fields are to be fixed width;
 - (ii) fields are to be space separated;
 - (iii) each field is to have a defined null value (for flagging of missing data) that is consistent with the format for that field;
 - (iv) there are to be no header or line header records;
 - (v) there are to be no blank records;
 - (vi) there are to be no non-numeric (e.g., *, /, \, or :) characters, except for those required for exponential formats (e.g., E or e);
 - (vii) each record is to end with a carriage return/line feed character.
- (c) Header files must be supplied for each ASCII column data file. The header file must include;
 - (i) brief description of processing steps applied;
 - (ii) any relevant calibration constants or parameters, including electromagnetic window times;
 - (iii) description of each field;
 - (iv) units of each field;
 - (v) sign convention where applicable;
 - (vi) column start and end position of each field;
 - (vii) null value for each field;
 - (viii) numeric format for each field (integer, real, decimal places etc.);
 - (ix) datum and projection for position fields.
- (d) The fields that uniquely identify each sample - project number, flight number, line number and fiducial number, must be contained in each record in all ASCII column files. These fields must be consistent between all files. These fields must be integers.
- (e) The project number will be the number as specified in the Request for Quotation, or as otherwise advised by The Customer . This project number may be different for different Survey localities flown under the one Contract.
- (f) Fiducial numbers should be integer values, must increase in regular increments and must not be repeated on any given flight line or tie line. Where a sample within a flight line or tie line is missing, at least the project number, flight number, line number and fiducial number must be present and other fields must be assigned their respective null value.

A1.2 - Gridded data format

- (g) Located data are to be gridded to a cell size of one-fifth of the minimum line spacing and in accordance with Section 12.12 of this Schedule.
- (h) Gridded data are to be delivered in ER Mapper binary raster grid format. The format must conform to the following criteria;
 - (i) data are to be stored as IEEE four or eight byte real data types;
 - (ii) the 'Datum' alias must be the Datum Alias;
 - (iii) the 'Projection' alias must be the Projection Alias;
 - (iv) the 'Units' alias must be appropriately assigned.

A1.3 - Digital data to be supplied

1. General

- (j) This Section 3 of Attachment 1 describes the minimum requirements for the digital data that are to be supplied. The complete list of digital data to be supplied is provided by the Supplier in Attachment 9.

2. Static Data

- 2.1. All static data collected pre- and post-flight will be recorded in a single spreadsheet table (the Static Data Table) in Microsoft Excel format or some other format agreed between the Customer and Supplier.
- 2.2. The static data will include at least the following:
- 2.2.1. repeat-point position data;
 - 2.2.2. gravity gradiometer Quiescent Noise data;
 - 2.2.3. gravimeter static test data;
 - 2.2.4. the date and time at which each of the above were recorded;
 - 2.2.5. a comment describing any event or observation that might have caused an unusual value in the above data.
- 2.3. The to-date and current spreadsheet will be provided to the Customer's Representative with the Weekly Report, or on reasonable request from the Customer's Representative.

3. Field Data

- 3.1. Field Data shall be delivered as point located data. It is expected that these data will have at least been checked and edited/cleaned for quality control purposes during the data acquisition phase.
- 3.2. Gravimetric, gravity gradiometric data and any ancillary data shall each be delivered in separate data sets.
- 3.3. Each and every delivered Field Data set shall contain at least the common channels listed in Table A1.1.
- 3.4. The gravity gradiometric data set that are to be supplied as Field Data shall contain at least the channels listed in Table A1.2.
- 3.5. The gravimetric data set that are to be supplied as Field Data shall contain at least the channels listed in Table A1.3.

Table A1.1 Common channels for gravimetric and gradiometry Field Data.

Field	Format	Units	Comment
Customer Project Number	I6		integer number
Supplier Job Number	I7		integer number
Flight Number	I4		integer number
Line Number	I10		integer number
Fiducial	I8		integer number
Local date	I9		yyyymmdd
Local time	F10.2	sec	Secs after local midnight
UTC Time	F16.3	sec	Coordinated Universal Time in seconds
Bearing	F8.1	deg	East of North

Table A1.1 Common channels for gravimetric and gradiometry Field Data.

Longitude	F13.7	deg	relative to the Position Datum
Latitude	F13.7	deg	relative to the Position Datum
Height	F8.2	m	relative to the Height Datum
Easting	F11.2	m	in Projected Coordinates
Northing	F11.2	m	in Projected Coordinates
Clearance	F8.2	m	Vertical separation between Aircraft and ground surface features
Turbulence	F7.3	m s ⁻²	Vertical acceleration filtered in [0.3, 1.0] Hz
GNSS Satellites	I4		number of GNSS satellites
PDOP	F6.1		dilution of precision
VDOP	F6.1		vertical dilution of precision
HDOP	F6.1		horizontal dilution of precision

Table A1.2 Gravity gradiometric channels for Field Data.

Field	Form at	Units	Comment
raw gravity gradients	F9.2	eotvos	Raw cross and in-line gravity gradients for each complement. Post- PMC and demodulation; pre- levelling, terrain correction, transformation and enhancement
gravity gradient noise	F9.2	eotvos	Noise estimate per gravity gradient channel; difference noise (dual-complement) or in-line sum (otherwise)
raw rotated gravity gradients	F9.2	eotvos	cross and in-line gravity gradients rotated into the Coordinate Frame. 3-axis systems only.
bias-corrected rotated gravity gradients	F9.2	eotvos	3-axis systems only.
DTM	F10.2	m	terrain height below the Aircraft; if available.
TC1	F9.2	eotvos	Terrain correction for each gravity gradient at 1.0 t m ⁻³ terrain density; if available.
CarouselAngle	F6.1	deg	angular position of platform; 3-axis systems only.
VertVelocity	F8.3	m s ⁻¹	Vertical velocity from GNSS; 3-axis systems only.
Status	I3		Any flag or indicator of data quality or useability supplied by Supplier.

Table A1.3 Gravimetric channels for Field Data.

Field	Forma t	Units	Comment
total gravimeter acceleration	F15.9	m s ⁻²	Unfiltered in data processing.
total navigation acceleration	F15.9	m s ⁻²	Computed from the navigation data. Unfiltered in data processing.

Table A1.3 Gravimetric channels for Field Data.

raw gravity	F15.9	m s ⁻²	Total gravimeter acceleration minus total navigation acceleration. Unfiltered in data processing.
Static correction	F9.3	um s ⁻²	Drift correction based on static measurements and applied to raw gravity
GRS80 Normal gravity	F9.3	um s ⁻²	Correction applied to raw gravity
Eotvos correction	F9.3	um s ⁻²	Correction applied to raw gravity
Atmospheric correction	F9.3	um s ⁻²	Correction applied to raw gravity
Free air correction	F9.3	um s ⁻²	With respect to the Height Datum. Correction applied to raw gravity.
raw free air anomaly	F9.3	um s ⁻²	Computed from raw gravity; unfiltered in data processing.
Filtered free air anomaly	F9.3	um s ⁻²	at agreed filter length(s)
Terrain-corrected vertical gravity	F9.3	um s ⁻²	at agreed filter length(s)
Terrain correction value applied	F9.3	um s ⁻²	A standard density value of 2.67 t m ⁻³ should be used for the terrain corrections.
Status	I3		Any flag or indicator of data quality or useability supplied by Supplier.

4. Final Data

- 4.1. Final Data shall be delivered as point located data and as gridded data.
- 4.2. Point located gravimetric data, gravity gradiometric data and laser scanner data (if acquired) shall be delivered in separate data sets.
- 4.3. The final processed point located gravimetric and gravity gradiometric Final Data sets shall contain at least the common channels listed in Table A1.4.
- 4.4. The final point located gravity gradiometric data shall contain at least the channels listed in Table A1.5.
- 4.5. The final processed point located gravimetric digital data shall contain at least the channels listed in Table A1.6.
- 4.6. The final point located laser scanner data shall contain the fields agreed between Customer and Supplier and listed in Attachment 9.

Table A1.4 Common channels for gravimetric and gradiometry Final Data.

Field	Format	Units	Comment
Customer Project Number	I6		integer number
Supplier Job Number	I7		integer number
Flight Number	I4		integer number
Line Number	I10		integer number
Fiducial	I8		integer number
Local date	I9		yyyymmdd
Local time	F10.2	sec	Secs after local midnight
UTC Time	F16.3	sec	Coordinated Universal Time

Table A1.4 Common channels for gravimetric and gradiometry Final Data.

Bearing	F8.1	deg	East of North
Final Longitude	F13.7	deg	relative to the Position Datum
Final Latitude	F13.7	deg	relative to the Position Datum
Final Aircraft Height	F8.2	m	Height of Aircraft Reference Pt relative to Height Datum
Final Easting	F11.2	m	in Projected Coordinates
Final Northing	F11.2	m	in Projected Coordinates
Final Ground Clearance	F8.2	m	Vertical separation between Aircraft and ground from ranging sensor(s)
Final Ground Elevation	F8.2	m	AHD

Table A1.5 Final processed gravity gradiometric data.

Field	Format	Units	Comment
Final Data Surface	F11.3	...	height at which processed gravity gradient data are calculated;
DTM & bathymetry	F8.2	m	Terrain height and bathymetric depth (sampled from final grids)
Turbulence	F7.3	m/s/s	Vertical acceleration filtered in [0.3, 1.0] Hz
levelled free-air measured gradients	F9.2	eotvos	inline and cross gradients for each complement
gradient noise estimate(s)	F9.2	eotvos	Noise estimate per gravity gradient channel; difference noise (dual-complement) or in-line sum (otherwise)
Terrain correction gradients	F9.2	eotvos	Calculated from DTM at 1.0 t m ⁻³ density
Bathymetric correction gradients	F9.2	eotvos	Calculated from bathymetry at 1.0 t m ⁻³ density
levelled free-air gradients	F9.2	eotvos	all components of the tensor in the Coordinate Frame representation
levelled terrain-corrected gradients	F9.2	eotvos	corrected for terrain and bathymetry; all components of the tensor in the Coordinate Frame representation
levelled free-air gravity	F9.2	um s ⁻²	calculated from gradients
levelled terrain-corrected gravity	F9.2	um s ⁻²	calculated from gradients
levelled free-air full spectrum gravity and gravity gradients	F9.2	um s ⁻²	all tensor components and vertical gravity calculated by merging gravity gradiometer and gravimeter data
levelled terrain-corrected full spectrum gravity	F9.2	um s ⁻²	all tensor components and vertical gravity calculated by merging gravity gradiometer and gravimeter data

(k)

Table A1.6 Final processed gravimetric data.

Field	Format	Units	Comment
total gravimeter acceleration	F15.9	m s ⁻²	Unfiltered in data processing.
total navigation acceleration	F15.9	m s ⁻²	Computed from the navigation data. Unfiltered in data processing.

Table A1.6 Final processed gravimetric data.

raw gravity	F15.9	m s ⁻²	Total gravimeter acceleration minus total navigation acceleration. Unfiltered in data processing.
Static correction	F9.3	um s ⁻²	Drift correction based on static measurements and applied to raw gravity
GRS80 Normal gravity	F9.3	um s ⁻²	Correction applied to raw gravity
Eotvos correction	F9.3	um s ⁻²	Correction applied to raw gravity
Atmospheric correction	F9.3	um s ⁻²	Correction applied to raw gravity
Free air correction	F9.3	um s ⁻²	With respect to the Height Datum. Correction applied to raw gravity.
raw free air anomaly	F9.3	um s ⁻²	Computed from raw gravity; unfiltered in data processing.
Filtered free air anomaly	F9.3	um s ⁻²	at agreed filter length(s)
Terrain-corrected vertical gravity	F9.3	um s ⁻²	Full Bouguer and terrain-corrected at agreed filter length(s)
Terrain correction value per unit density	F9.3	um s ⁻²	Full Bouguer and terrain correction. A density value of 1.0 t m ⁻³ should be used for the terrain corrections.
Status	I3		Any flag or indicator of data quality or useability supplied by Supplier.

Attachment 2 - Calibrations and Tests

A2.1. Calibration Reporting

- (i) Calibrations and test required before the commencement of the Survey shall be reported in a Commencement Report which will include details of all procedural and processing steps and the results.
- (j) Calibrations and tests performed after the commencement of Survey acquisition shall be reported as required during the Survey as part of the Daily Report. All procedural and processing steps and the results will be reported in detail in the Final Report.

A2.2. Ranging sensor calibration

1. Altimeter linearity checks

- 1.1. Radar and laser altimeter linearity will be verified before commencement of Survey operations by flying over a level airstrip, or similar flat surface, at Aircraft terrain clearances of 30, 60, 90, 120, 150 and 180 m.
- 1.2. Measured averages for each height-pass from the laser altimeter(s), radar altimeter, and all GNSS instruments will be compared in regression cross-plot form to determine consistency between the instruments. Any non-linearity or offsets between the instruments shall be corrected for in the data processing.
- 1.3. All results of the altimeter calibrations are to be compiled for inspection by Customer's Representative before commencement of the Survey. The results will also be included in the Final Report.

2. Laser altimeter

- 2.1. Laser altimeter calibration constants will be determined prior to the commencement of the Survey by the methods described here. These calibration constants are to be used in the data processing to correct for effects caused by the laser altimeter lens not pointing vertically downward. The calibrations constants to be determined are:
 - 2.1.1. the laser altimeter's pointing roll and pitch angle offsets relative to the Aircraft's orientation when it is horizontal, and
 - 2.1.2. the horizontal and vertical offsets between the laser altimeter and the Aircraft's centre of rotation.

3. Laser scanner

- 3.1. Laser scanner calibration constants will be determined prior to the commencement of the Survey by the methods described here. These calibration constants are to be used in the data processing. The calibrations constants to be determined are:
 - 3.1.1. the laser scanner's pointing roll and pitch angle offsets relative to the Aircraft's orientation when it is horizontal, and
 - 3.1.2. the horizontal and vertical offsets between the laser scanner and the Aircraft's centre of rotation.

A2.3. Gravimeter calibration and tests

1. Local gravity base and tie to Australian Fundamental Gravity Network

- 1. The aircraft will be parked in the same spot(s) each day for the purpose of static testing. A local reference point will be marked at the aircraft parking spot(s) as close as practically possible below the actual location of the gravimeter sensors in the aircraft. The same procedures will be followed for each operational base from which the aircraft operates. The assistance of the Customer may be requested for the reference point tie(s) to the AFGN station.

2. Pre-survey test line

- 2.1. To demonstrate the current accuracy, repeatability and resolution of the airborne gravity system, the Supplier must fly a pre-survey test line before survey data acquisition commences with the aircraft and gravity system to be used on the survey in standard survey configuration.
- 2.2. The test line must be at least 40 km in length along a path with existing ground gravity data with a station spacing no more than 2.5 km.
- 2.3. The test will consist of four passes, two in each direction.
- 2.4. Data from the test line passes must be processed in the same way as proposed for the survey.
- 2.5. A complete dataset must be submitted to demonstrate the range of parameters recorded and their compliance with the specifications. The Customer will respond within a period of 2 working days from receipt of the data by the Customer. Survey data acquisition must not start until Customer advises the Supplier that it is approved to do so. Any data collected prior to approval will be at the Supplier's own risk.

Line Number	West end coordinates		East end coordinates		Bearing	Altitude above ground level
	MGA94 Zone 50 East	MGA94 Zone 50 North	MGA94 Zone 50 East	MGA94 Zone 50 North		
6468N	485 000	6 468 000	525 000	6 468 000	090/270	Survey configuration

The preferred test line is located at the Kauring Gravity Test Range as follows:

- 2.6. The Supplier is responsible for making all arrangements for test flights over the Kauring Gravity Test Range including notification to land-holders and other air space users. Location and access protocols for the test range are available at <http://www.ga.gov.au/about/projects/resources/kauring>.
- 2.7. An alternative test line may be substituted if agreed by the Customer. Ground data for the alternative line must be supplied together with the airborne test data.

3. Daily static test for gravimeter accuracy and precision

- 3.1. Before and after each flight, the consistency of the measured gravity will be confirmed by recording gravity data at the same fixed location on the ground for a period of 10 minutes or more while the aircraft engines are not running. The location of the aircraft must be determined by processing the GNSS data acquired at the same time as the gravity data. Contact with the aircraft during acquisition of static gravity and navigation data will be avoided to minimize accelerations caused by aircraft motion. The results of each static test will be included in the Daily Report, and the results of all static tests will be tabulated and presented in the Final Report, as deviations from the local gravity value at the base. Gravity and navigation processed static readings must be reported to the Customer within 24 hours of every completed flight.

4. In-survey repeat line

- 4.1. Traverses along a repeat line will be flown in normal survey configuration as described below.
- 4.2. The repeat line will be at least fifty (50) kilometres in length.
- 4.3. The location of the repeat line shall be chosen in Consultation between the Supplier and the Customer's Representative. As a guide, the repeat line will be located:
 - 4.3.1. inside the Survey Area;
 - 4.3.2. so that it does not cause regular disturbance to local residents or livestock;
 - 4.3.3. in an operationally efficient position for the Supplier;
 - 4.3.4. in an area that produces minimal cultural noise sources;
 - 4.3.5. in an area of topographic relief that is representative of the survey area as a whole;
 - 4.3.6. in an environment where the mass density structure is known or considered to be representative of the Survey Area.
- 4.4. The repeat line is to be flown at least:
 - 4.4.1. once every day for the first three successful production flying days, then thereafter,

- 4.4.2. once every four days of successful production flying up to the maximum number of passes specified in **Section 1.5**, and
- 4.4.3. after modification to or maintenance on the instruments that impact the measurements; e.g., altimeters, attitude sensors, navigation system, or gravimeter instrument, in this instance at the Supplier's cost.
- 4.5. On each occasion, the repeat line will be flown in standard Survey operating mode and as close as possible to the nominal terrain clearance.
- 4.6. The repeat line should be flown alternately in each direction such that there is approximately the same number of passes in each direction.
- 4.7. All repeat line data including lines acquired but not chargeable to the Customer, are to be processed and supplied in the same manner as the Survey line data.
- 4.8. The Supplier must demonstrate the repeatability or otherwise of its system through appropriate graphical presentations of comparisons between responses recorded each time the repeat line is flown. This will be achieved via profile plots in the field.
- 4.9. Repeat line Data are a primary part of the dataset and accordingly the digital Data must be delivered in accordance with relevant clauses of the contractual specifications.
- 4.10. Results and analysis of the calibrations and graphical representation of system repeatability must be included in the Final Report.

A2.4. Gravity gradiometer calibration and tests

1. Sensor calibrations and tests

- 1.1. The gravity gradiometer must be calibrated and tested according to the manufacturer's recommendations before the start of the Survey.
- 1.2. If any maintenance or repair is performed on the gravity gradiometer during the course of the Survey, the calibrations and tests must be completed according to the manufacturer's recommendations before acquisition re-commences.
- 1.3. The Final Report must include the dates of the last calibration and test before the Survey as well as the dates of any such during the Survey. The result ("pass" or "fail") of each must also be included.

2. Gravity test range survey

- 2.1. To demonstrate the current accuracy, repeatability and resolution of the airborne gravity gradiometer system, the Supplier must provide Final Data and a Final Report from a survey flown over a gravity test range suitable for airborne gravity gradiometry. The Test Range Data and Report must be provided as part of the Commencement Report.
- 2.2. The test range survey should be flown with the aircraft and gravity gradiometer system to be used on the Survey and in standard survey configuration, unless the Supplier can demonstrate that the aircraft and system to be used will deliver the same level of performance or better.
- 2.3. The test range survey must have a line spacing and survey height above ground no greater than proposed for the Survey and must cover an area of length and width both no smaller than the greater of 5 km and 50 times the line spacing.
- 2.4. The test range area must have ground gravity data sampled at an average sample spacing less than 100 m at positions mapped to a horizontal accuracy better than 1 metre and a vertical (height) accuracy better than 0.1 metre. The RMS trueness of the ground gravity shall be better than $0.2 \text{ } \mu\text{m s}^{-2}$.
- 2.5. Data from the test range survey must be processed in the same way as proposed for the survey.
- 2.6. A complete dataset must be submitted to demonstrate the range of parameters recorded and their compliance with the specifications. The Customer will respond within a period of 2 working days from receipt of the data by the Customer.
- 2.7. The preferred test range is located at the Kauring Gravity Test Range in Western Australia (Howard et al., 2010). Location and access protocols for the test range are available at <http://www.ga.gov.au/about/projects/resources/kauring>.
- 2.8. An alternative test range may be substituted if agreed by the Customer. Ground data for the alternative range must be supplied together with the airborne test data.

- 2.9. The Supplier is responsible for making all arrangements for test flights over the Kauring Gravity Test Range or any alternative test range including notification to land-holders and other air space users.

3. Self-gradient calibration

- 3.1. A self-gradient calibration must be performed before the Survey commences. The coefficients from the calibration are to be used in self-gradient corrections.
- 3.2. If the position of the gravity gradiometer within the Aircraft is changed, or any significant changes are made to the mass distribution within the Aircraft, during the course of the Survey, then a new self-gradient calibration must be performed and the new coefficients used for self-gradient corrections.
- 3.3. The Final Report must include the date of the last self-gradient calibration before the start of the Survey and the dates of any such performed during the Survey.

4. Quiescent tests

- 4.1. Before each survey flight a measurement of the gravity gradiometer Quiescent Noise must be performed.
- 4.2. Quiescent Noise measurements should record the total system signal power per complement within a representative frequency band while the Aircraft is parked and undisturbed. The measurement should be made over a duration of at least 15 minutes.
- 4.3. If the Quiescent Noise is unusually high and there is no evidence of external disturbance, then no survey flight shall be attempted until a low Quiescent Noise measurement is obtained.
- 4.4. The Daily Report must include the Quiescent Noise, in units of eotvos squared per radian per second, and the frequency band over which it was measured, for each survey flight.

Attachment 3 - Measurement accuracy and resolution

A3.1 Accuracy, trueness and precision

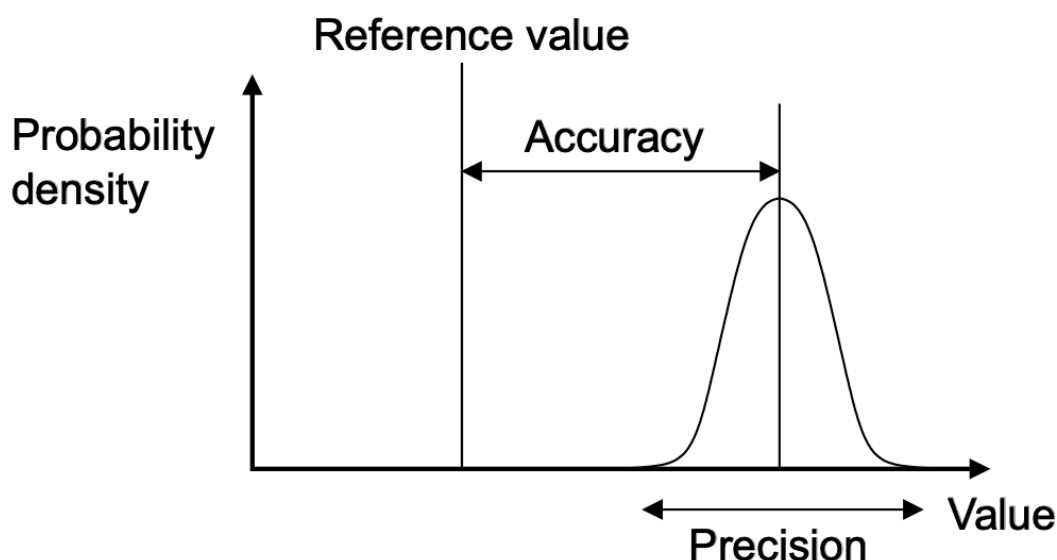
The definitions of accuracy, trueness and precision are taken from:

ISO 5725-6:1994(en) — Accuracy (trueness and precision) of measurement methods and results – Part 1: General principles and definitions.

(<https://www.iso.org/obp/ui/#iso:std:iso:5725:-1:ed-1:vi:en>)

ISO 5725 uses two terms "trueness" and "precision" to describe the accuracy of a measurement method. "Trueness" refers to the closeness of agreement between the arithmetic mean of a large number of test results and the true or accepted reference value. "Precision" refers to the closeness of agreement between test results.

The general term accuracy is used in ISO 5725 to refer to both trueness and precision.



A3.2 Definitions

accuracy	The closeness of agreement between a test result and the accepted reference value. The term accuracy, when applied to a set of test results, involves a combination of random components and a common systematic error or bias component.
trueness	The closeness of agreement between the average value obtained from a large series of test results and an accepted reference value. The measure of trueness is usually expressed in terms of bias.
bias	The difference between the expectation of the test results and an accepted reference value. Bias is the total systematic error as contrasted to random error. There maybe one or more systematic error components contributing to the bias. A larger systematic difference from the accepted reference value is reflected by a larger bias value.
precision	The closeness of agreement between independent test results obtained under stipulated conditions. Precision depends only on the distribution of random errors and does not relate to the true value or the specified value. The measure of precision is usually expressed in terms of imprecision and computed as a standard deviation of the test results. Less precision is reflected by a larger standard deviation.

A3.3 Instrument recording resolution

The resolution of an instrument is the smallest detectable change in measurement (often implied by the least significant digit of the reported value, e.g. if data in um/s^2 are reported to 2 decimal places, the resolution is assumed to be 0.01 um/s^2).

A3.4 Spatial resolution

The *spatial resolution* of a set of discrete, equally spaced measurements is the distance between two completely independent measurements.

The *spatial wavelength resolution* of the set is twice that distance.

When data are over sampled — as is typically the case in high-frequency data recording — the spatial resolution is larger than the sample interval and is typically dependent on the length of any spatial filtering applied to the data.

The above definitions notwithstanding, the Customer accepts for the convenience of discussion and reporting an alternate definition of the spatial wavelength resolution of a filtered dataset as the wavelength that has a gain of 0.5 (50% pass) in a filter gain spectrum. Under this alternate definition, the spatial resolution is then half the spatial wavelength resolution.

In order to properly indicate how the data between the pass band (100% pass) and the stop band (0% pass) are affected by a given spatial filter, it will be necessary for the characteristics of the filter to be fully specified and, preferably, supported by a gain spectrum diagram.

Given the plurality of definitions, it will be a requirement that (1) the spatial resolution and spatial wavelength resolution values for both of the definitions be provided and (2) there be an explicit and consistent use of whichever one of the two definitions are used in any survey reports.

Attachment 4 - Coverage

Customer and Supplier to provide details of Survey coverage as shown below.

A4.1 Positioning

- (l) The Aircraft is to be flown along the planned lines and at the planned height to the requirements of Table A4.1.
- (m) Data along line segments that do not meet the flying requirements of Table A4.1 are Unacceptable Data unless it can be reasonably demonstrated that such flying would not be safe or that the deviation is required by civil aviation requirements.

Table A4.1 Positioning

Data type	Maximum difference	For a distance
Difference in horizontal position from plan	20 m	< 1000 m
Difference in height from planned height	10 m	< 1000 m
Difference in height between traverse lines and tie lines	10 m	at cross-over point
Difference in heading from plan	1 degrees	< 1000 m
Variation in survey ground speed	10 % of nominal (Attachment 5, Table 5.3)	< 1000 m

A4.2 Gaps in data

1. Gaps in the data for any line must not exceed the limits shown below for each data type. Isolated errors, spikes or gaps in the data within this tolerance level may be corrected by interpolation up to the maximum number of gaps shown for any given flight line, providing no similar data gaps exist on nearby sections of adjacent flight lines. Gaps that exceed the limits shown in the table constitute Unacceptable Data.

Table A4.2 Data gaps

Data type	Maximum gap	Maximum number
Gravimetry	2.0 sec	2
Gravity gradiometry	0.5 sec	2
Navigation	0.5 sec for AGG, 2.0 sec for AG	2
Ranging	5. sec for AGG, 2.0 sec for AG	2

A4.3 Re-flight line segments

1. The Data from a re-flight line segments that does not satisfy the requirements of Table A4.3 will be deemed Unacceptable Data.

Table A4.3 Re-flight line segments

Aspect	Specification, treatment, tolerances
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Table A4.3 Re-flight line segments

Minimum length	10 km for AGG; 40 km for AG
Minimum overlap with accepted traverse line	To intersect at least 2 tie lines and an overlap with the accepted traverse > Data Filter Extension + Run-in / Run-out.
Minimum overlap with accepted tie line	Data Filter Extension + Run-in / Run-out.
Re-flight direction	The same as the line segment that is being replaced.

Attachment 5 - Aircraft Specifications

A5.1 Aircraft

Supplier to provide details of all aircraft to be used for the Survey. The tables gives examples of the type of information required, but can be altered as required. One pair of tables should be provided for each aircraft.

Table A5.1 Specification for Aircraft

Registration	
Make and Model	
Ownership	
Serial No	
Date of Manufacture	
Engines	
Propellers	
Dimensions	
Fuel	
Fuel Capacity	
Endurance	
Total Airframe Time	
Seating	
Flight Instruments	
Nominal ground speed (metres/sec)	

Table A5.2 Engine performance of Aircraft

Temperature (deg C)	Elevation (m)	Rate of Climb (ft/min)
0	0	
20	0	
40	0	
0	2000	
20	2000	
40	2000	

A5.2 Data acquisition rate

Supplier to provide details for the Survey.

Table A5.3 Data acquisition rate

Aspect	Supplier to complete
Survey ground speed	_____ m/s
Aircraft endurance	_____ hours per flight
Typical daily production rate	_____ km per day
Expected average production rate	_____ km per day over the course of the survey

Attachment 6 – Equipment Specifications

Supplier to provide details of the equipment to be used for the Survey. The equipment must be able to meet the requirements specified here and elsewhere in this Deed. All information should be provided before contract signing unless otherwise agreed between Customer and Supplier.

A6.1. General configuration

Table A6.1 Position of the instruments relative to the GPS antenna.

Instrument	In-line X (m) flight dirn. +ve	Transverse Y (m) left side +ve	Vertical Z (m) upward +ve
gravimeter centre	0.0	0.0	0.0
gravity gradiometer centre			
laser scanner			
Laser altimeter 1			
Laser altimeter 2			
Aircraft radar altimeter			
Other instruments as required			

A6.2. Data acquisition system

Details of the data acquisition and recording system. This should include the types of files recorded and the method of synchronising data from each instrument, and fiducial system.

A6.3. Navigation and positioning

1. Navigation

Details of the in-flight navigation system and pilot display.

2. Aircraft GNSS

Details of aircraft GNSS used for navigation.

Table A6.2 Aircraft navigation GNSS.

Make and model	
Recording interval	
Resolution	
Accuracy	

Table A6.2 Aircraft navigation GNSS.

Mounting position on aircraft	
Real time differential correction method	
Post-processed differential corrections method	
Data recorded	latitude, longitude, height, time

3. Base station GNSS

Details of GNSS base station.

Table A6.3 Base station GPS specifications.

Make and model	
Recording interval	
Resolution	
Accuracy	
Typical location	
Method of establishing position	
Data recorded	latitude, longitude, height, time

A6.4. Ground ranging

Details of any altimeters and ranging sensors to be provided in the Tables below.

1. Radar Altimeter

Table A6.4 Radar altimeter specifications.

Aspect	Specification, treatment, tolerances	Supplier info
Radar altimeter	Make and model	
Recording frequency	≥ 5 Hz	
Resolution	≥ 0.1 m	
Absolute accuracy	m	
Relative accuracy	%	
Range	m	
Mounting position	relative to the GNSS antenna in m	

2. Laser Altimeter

Table A6.5 Laser altimeter specifications.

Aspect	Specification, treatment, tolerances	Supplier info
Laser altimeter	Make and model	
Recording frequency	≥ 10 Hz	
Resolution	≥ 0.05 m	
Accuracy	m	
Range	m	
Mounting position	relative to the GNSS antenna in m	

3. Laser Scanner

Table A6.6 Laser scanner specifications.

Aspect	Specification, treatment, tolerances	Supplier info
Laser scanner	Make and model	
Recording frequency	Hz	
Resolution	m	
Accuracy	≤ 0.1 m	
Range	m	
Swath angle	$\geq \pm 40$ degrees	
Mounting position	relative to the GNSS antenna in m	

A6.5. Gravity Gradiometer Specifications

1. Gravity gradiometer system

Details of the survey gravity gradiometer to be entered in Table A6.7. The table should be repeated for each separate gravity gradiometer to be used.

- 1.1. Current commercially available airborne gravity gradiometers are all of the rotating sensor type and consist of either one or three gravity gradiometer instruments (GGIs). Each GGI contains a rotor with either one or two complements of four accelerometers mounted with their sensitive axis tangential to the rotor and at a common distance from the rotor spin axis. This distance on a dual-complement GGI is double that on a single-complement GGI (Difrancesco, 2001). The spin axes on a triple axis system are all orthogonal. Three arrangements are available:
 - 1.1.1. single, near-vertical axis, dual-complement;
 - 1.1.2. triple axis, single-complement; and
 - 1.1.3. triple axis, dual-complement.
- 1.2. The gravity gradiometer used in the survey will be the one identified in Table A6.7.

Table A6.7 Gravity gradiometer sensor specifications.

Aspect		Supplier Info
Gravity gradiometer	make & model	
Data sample rate	System acquisition rate in Hz	
Processed data sample rate	> 1 Hz	
Measurement resolution	smallest detectable change in measurement; < 0.05 E	
Mounting position	within the Aircraft	
Number of independent GGI axes	1 or 3	
Number of complements per GGI axis	1 or 2	

- 1.3. The gravity gradiometer's recording system must be able to provide a post-processing measure of the components of Earth's gravity gradient to at least the tolerances specified in Section 5.2.

2. Noise Characteristics

- 2.1. These Noise Characteristics shall be used by the Customer when making its assessment of the technical suitability of the system for a specific survey and as a quality control check on data during the survey. The noise characteristics for the system are specified in Table A6.8(a) and (b). Note the provision under Section 13 of this Schedule (Continual Assessment) for the Supplier to provide up to date sample data.
- 2.2. Data failing to meet the tolerances specified in Table A6.8 are Unacceptable Data.
- 2.3. The following definitions are used in Table A6.8.
- 2.3.1. **Demodulation Filter** is the filter applied after demodulation to limit the signal bandwidth (sometimes called "the bandpass filter"). The characteristics of this filter must be clearly specified by the Supplier in Table A6.8.
- 2.3.2. **Raw Gradiometer Data** means digital gravity gradient data after Post Mission Compensation, demodulation and application of a Demodulation Filter but before any additional noise reduction, re-sampling or filtering processes.
- 2.3.3. **Measurement Bandwidth** is the bandwidth of the Demodulation Filter, specified by its 3dB frequency(ies).
- 2.3.4. **Survey Noise**: The RMS error per gravity gradient tensor component along a survey line segment over the Measurement Bandwidth. This should be calculated as the standard deviation of half the difference between two complements for dual-complement systems (and averaged over the tensor components), or, for a triple-axis, single-complement system, the sum of the three in-line components divided by the square root of three and averaged over the line segment. For triple-axis, dual-complement systems, the difference noise method is preferred. Calculations must be on Raw Gradiometer Data.
- 2.3.5. **Final Data Spatial Resolution**: The estimated Spatial Resolution of the delivered vertical gravity gradient data in metres.
- 2.3.6. **Final Data Spatial Wavelength Resolution**: The estimated Spatial Wavelength Resolution of the delivered vertical gravity gradient data in metres.
- 2.3.7. **Final Data Error**: The estimated RMS error, at the **Final Data Spatial Resolution**, in the delivered vertical gravity gradient data. The error estimate should be supported by an analysis of data collected over a gravity test range such as the R. J. Smith test range at Kauring, Western Australia. A reasonable analysis would include calculation of the RMS difference between final free air vertical gravity gradient values from airborne and upward-continued, vertically-differentiated free-air ground data over the same area, with both data sets having the same filter applied. Details of the analysis should be included by the Supplier as an Appendix to the Commencement Report.

Table A6.8 Gravity gradiometer measurement characteristics.

Aspect	Specification	Supplier Info
Static precision frequency range	The frequency range in units of Hz over which the Quiescent Noise measurements are made.	
Typical Quiescent Noise	The nominal Quiescent Noise expected.	
Demodulation Filter name	Filter name (eg Butterworth, Cosine, Chebyshev, etc) or description.	
Demodulation Filter frequencies	Filter key frequencies (eg 3db point for Butterworth, 100% cutoff for Cosine) in Hz.	
Survey mode precision bandwidth	The bandwidth, in Hz, used for the survey mode precision measurement. This should be the Measurement Bandwidth.	
Survey mode precision estimation method	Either dual-complement difference or in-line sum.	
Survey mode precision	Maximum allowable Survey Noise in E; Data with a higher Survey Noise will be deemed Unacceptable Data.	
Expected trueness in survey mode	The Final Data Error in units of E.	
Names of filter(s) for all deliverable gravity gradients	Filter names (Cosine, Chebyshev, etc) or descriptions.	
Key frequencies of filter(s) for all deliverable gravity gradients	Filter key frequencies (eg 3db point for Butterworth, 100% cutoff for Cosine) in Hz.	
Spatial resolution of deliverable gravity gradients	Final Data Spatial Resolution in metres.	
Spatial wavelength resolution of deliverable gravity gradients	Final Data Spatial Wavelength Resolution in metres.	
Level of turbulence below which these specifications are met	Turbulence is defined as the standard deviation, along a survey line, of the vertical acceleration over a bandwidth of [0.3, 10.0] Hz, in units of ms ⁻² .	

A6.6. Gravimeter Specifications

1. Gravimeter system

Details of the survey gravimeter to be entered in Table A6.9. The table should be repeated for each separate gravimeter to be used.

1.1. The gravimeter used in the survey will be the one identified in Table A6.9.

Table A6.9 Gravimeter specifications.

Aspect	Specification, treatment, tolerances	Supplier Info
Gravimeter	make & model	
Measurement resolution	smallest detectable change in measurement $\leq 0.1 \mu\text{m/s}^2$	
Data sample rate	System acquisition rate in Hz	

Table A6.9 Gravimeter specifications.

Processed data sample rate	Must be > 1 Hz	
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- 1.2. The gravimeter's recording system must be able to provide a post-processing measure of the vertical component of the Earth's gravity field to at least the tolerances specified in Section 6.2.

2. Noise Characteristics

- 2.1. These Noise Characteristics shall be used by the Customer when making its assessment of the technical suitability of the system for a specific survey and as a quality control check on data during the survey.
- 2.2. Data failing to meet the tolerances specified in Table A6.10 are Unacceptable Data. Where the specification applies to a test or repeat line, all the data from the relevant flight are Unacceptable Data.
- 2.3. The noise characteristics for the system are specified in Table A6.10. Note the provision under Section 13 of this Schedule (Continual Assessment) for the Supplier to provide up to date sample data.

Table A6.10 Gravimeter noise characteristics.

Aspect	Specification, treatment, tolerances	Supplier Info
Static trueness (closeness to value of Earth's field)	Difference between mean of readings at reference station and 'true' value at station in $\mu\text{m/s}^2$	
Static precision (repeatability of measurement)	Standard deviation of readings at reference station from daily gravimeter static test in $\mu\text{m/s}^2$	
Expected trueness in survey mode	Comparison with test line ground gravity data (Attachment 2, Section 3). Trueness is estimated as RMS of differences between free air anomaly values from airborne and ground data along coincident lines or over the same area, with both data sets having the same filter applied. RMS in $\mu\text{m/s}^2$ and filter length in metres to be supplied.	
Expected heading difference	RMS difference between gravimeter free air anomaly values from the test line flown in opposite directions in $\mu\text{m/s}^2$.	
Expected precision in survey mode (repeatability of measurement along flight line)	In $\mu\text{m/s}^2$ using a 100 s full-wavelength filter. Actual precision in survey mode will be determined from repeat line statistics as the standard deviation of differences between terrain-corrected vertical gravity values along each repeat line pass (Att. 4.3.3) and corresponding values from mean of all passes.	
Proposed filtering for deliverable free-air gravity	Deliverable free-air gravity is the result obtained after standard corrections and filtering of raw gravity data (see Att. 4.5).	
Filter duration	Proposed filter 100% pass frequency in seconds or seconds-equivalent if applied in the space domain.	

Table A6.10 Gravimeter noise characteristics.

Minimum nominal spatial resolution of deliverable free-air gravity	Nominal spatial resolution is defined as filter duration (from aspect 10) times the nominal ground speed in m/s (Att. 1, aspect 5.1) and in units of m.	
Along-line post-filtering gravity data acceptability	The RMS gravity residual intersection noise levels in $\mu\text{m/s}^2$ from control line intersection differences for an individual flight line, as calculated by intersection differences divided by the square root of 2. These calculations will be performed to data after the following processing steps have been performed: i) Application of a 100s full-wavelength line filter. ii) Application of a 0th order (i.e. one constant shift per line) levelling.	
Level of turbulence below which these specifications are met	Turbulence is defined as the standard deviation, along a survey line, of the vertical acceleration over a bandwidth of [0.3, 10.0] Hz and reported in ms^{-2} .	

A6.7. Other Equipment Specifications

Details of any other equipment (e.g., pressure gauge, thermometer, video flight path recorder) to be entered here.

Attachment 7 - Data Verification Facility

Supplier to provide details of the data facility used to verify that data have been acquired in accordance with the technical specification on a daily basis. The facility must be able to carry out the tasks in Section 4.1 of this Schedule at a minimum.

A7.1 Hardware

A7.2 Software

A7.3 E-mail and internet

Attachment 8 - Data Processing Methods

Supplier to provide full details of the processing methods used for each section before contract signing. Any filters used must be fully specified including details of filter type, shape and length. Any micro–levelling techniques that may be applied should be described.

A8.1 Navigation and position data processing

A8.2 Ranging data processing

A8.3 DTM data processing

A8.4 Gravimetric Data processing

A8.5 Gravity Gradiometric data processing

A8.6 Joint Gravimetric and Gravity Gradiometric Data processing

A8.7 Ancillary Data processing

Attachment 9 - Deliverables

Supplier to provide details of the all deliverables before contract signing. The deliverables must comply with the requirements of the Deed, including those of Attachment 1.

A9.1 General

A9.2 Reports

A9.3 Digital data

- (a) General**
- (b) Line data**
- (c) Grid data**
- (d) Images**